



Overview of AI technologies within **CLIMINVEST**



Co-funded by
the European Union

Climinvest Introduction
Richard McCreadie (University of Glasgow)

From User Journey to Technology Stack

If you recall from earlier, I presented the user-journey for a project developer transitioning from an initial idea to a final investment case they could pass to a funder:



The problem with this view is it masks the practical complexity in implementing these tools, instead we should look at the project in terms of software layers

Web Apps

User Facing services, that have some form of front-end component that the user interacts with

Tools / 'Skills' / Models

Back-end services that provide Application Programming Interfaces that perform some task

Data Services

Services that store, manage and serve underlying data, typically databases, search engines or other structured data stores

The Tech Stack; Starting with the Data

We are going to work bottom-up; data is what drives all of the services that we can provide, there are four main data sources

- **Project Digital Model**: This is the central store for information about a green project in a structured form
- **Project Search Engine**: This is a passage store containing (AI-enriched) extracts of content about each project that are easier to consume by AI agents
- **Web Artifact Store**: This holds additional relevant content that has been collected via semi-automated research activities
- **Knowledge Base**: This is a general text information store that contains relevant cross-project information, such as laws and regulations

DATA



Project Digital Model



Project Search Engine



Web Artifact Store



Knowledge Base

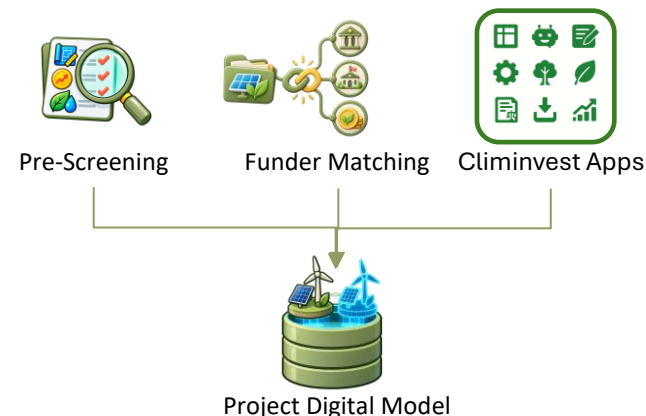
The Project Digital Model

One of the core tenets of Climinvest is the idea of being **Digital-First**

- **Pre-Screening** and **Funder Matching** and various **Climinvest apps** store the information they collect or create

Why?

- Recall that the goal is to make the creation of high-quality business cases as easy as possible; while we cannot fully rely on AI systems to complete this task, but we can have them help
- So, we need to collect information about the developer's project idea in a way that AI agents can understand



So we create a digital twin of each project called the **Project Digital Model**

...which is used by AI tooling later

What's in a Project?



Technology Stack

Remember how investment cases we need to create are complex documents with lots of information?



PROJECT / PROGRAMME SUMMARY
GREEN CLIMATE FUND FUNDING PROPOSAL | PAGE 1 OF 45



A.1. Brief Project / Programme Information											
A.1.1. Project / programme title	Catalyzing private investment in sustainable energy in Argentina - Part 1										
A.1.2. Project or programme	Project										
A.1.3. Country (ies) / region	Argentina										
A.1.4. National designated authority (ies)	National Directorate of Programs with International Credit Organizations (DNP/OIC)										
A.1.5. Accredited entity	Inter-American Development Bank										
A.1.5.a. Access modality	☐ Direct ☒ International										
A.1.6. Executing entity / beneficiary	Executing Entity: Inter-American Development Bank and Inter-American Investment Corporation Beneficiary: various renewable energy projects										
A.1.7. Project size category (Total investment, million USD)	☐ Micro (<10) ☐ Small (10<x<50) ☐ Medium (50<x<250) ☒ Large (>250)										
A.1.8. Mitigation / adaptation focus	☒ Mitigation ☐ Adaptation ☐ Cross-cutting										
A.1.9. Date of submission	September 26 th , 2016										
A.1.10. Project contact details	<table border="1"> <tr> <td>Contact person, position</td> <td>Gian Franco Carassale - Lead Investment Officer Alfredo Idiarte - Advisory Services Team Gloria Visconti - Climate Change Lead Specialist</td> </tr> <tr> <td>Organization</td> <td>Inter-American Development Bank gianfranco@iadb.org alidiarte@iadb.org gloria@iadb.org</td> </tr> <tr> <td>Email address</td> <td></td> </tr> <tr> <td>Telephone number</td> <td>+1 202 823 3360</td> </tr> <tr> <td>Mailing address</td> <td>1300 New York Ave NW, Washington DC 20577</td> </tr> </table>	Contact person, position	Gian Franco Carassale - Lead Investment Officer Alfredo Idiarte - Advisory Services Team Gloria Visconti - Climate Change Lead Specialist	Organization	Inter-American Development Bank gianfranco@iadb.org alidiarte@iadb.org gloria@iadb.org	Email address		Telephone number	+1 202 823 3360	Mailing address	1300 New York Ave NW, Washington DC 20577
Contact person, position	Gian Franco Carassale - Lead Investment Officer Alfredo Idiarte - Advisory Services Team Gloria Visconti - Climate Change Lead Specialist										
Organization	Inter-American Development Bank gianfranco@iadb.org alidiarte@iadb.org gloria@iadb.org										
Email address											
Telephone number	+1 202 823 3360										
Mailing address	1300 New York Ave NW, Washington DC 20577										

A.1.11. Results areas (mark at that apply)	
Reduced emissions from:	
☒	Energy access and power generation (E.g. on-grid, mini-grid or off-grid solar, wind, geothermal, etc.)
☐	Low emission transport (E.g. high-speed rail, road bus system, etc.)
☐	Buildings, cities and industries and appliances (E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.)
☐	Forestry and land use (E.g. forest conservation and management, agroforestry, agricultural irrigation, water treatment and management, etc.)
Increased resilience of:	
☐	Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change - diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.)
☐	Health and well-being, and food and water security (E.g. climate-resilient crops, efficient irrigation systems, etc.)



FINANCING / COST INFORMATION
GREEN CLIMATE FUND FUNDING PROPOSAL | PAGE 3 OF 45



B.1. Description of Financial Elements of the Project / Programme							
Please provide a description of how the choice of financial instrument(s) will overcome barriers and achieve project objectives, and leverage public and/or private finance.							
The use of GCF investment resources will be aimed at helping structure and complete the long-term debt packages required for projects to be financially viable at competitive prices. For this, GCF resources will be made available in two modalities: 1) long-term project debt, matching the tenor that the IDBG will provide. 2) refinancing guarantees, which would help commercial lenders and commercial debt investors extend the tenor they can offer to those required by projects, otherwise generally restricted to 5-7 years. The guarantees would allow sponsors to accept financing that give these lenders the possibility to exercise early repayment options, by guaranteeing the refinancing of such loans. This instrument thus has the potential to crowd in commercial lenders and investors that would otherwise not be able to extend suitable terms and participate in the financing to deliver the volumes needed by the ambitious targets of the RenovAr public tendering program. Through these two GCF instruments, IDBG expects to help leverage additional, adequately termed debt and timely close projects, to reduce the incidence of drop outs (awarded projects that are never built) that has been so high in previous tenders (just about 15% of the awarded capacity built in the previous renewable energy tender, in 2010). The proposed USD 130M from the GCF would help mobilize investment for a total of about USD 650M, and support about 400 MW, which is equivalent to about 40% of the capacity originally targeted by the first tender under RenovAr. If achieved, the GCF Program will have helped provide a meaningful contribution to the success of this government initiative with long-term demonstration and capacity building effects. This would help pave the way for the significant replication and scale-up required in the mid and long term to achieve Law 27.191's ambitious renewable penetration goals.							
Component	Sub-component (if applicable)	Amount (for entire project)	Currency	Amount (for entire project)	Local currency	GCF funding amount	Currency of disbursement to recipient
Component 1: Financing of renewable energy power projects	Sub-component 1.1: Financing of renewable energy power projects	650	million USD (\$)			130 million USD	USD
Component 2: Enhancing policy, planning, regulation and project financing capacity	Sub-component 2.1: Support for sustainable development, operation and financing of renewable energy projects	1.0	million USD (\$)			1.0 million USD	USD



EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA
GREEN CLIMATE FUND FUNDING PROPOSAL | PAGE 26 OF 45



Please describe financial viability in the long run beyond the Fund intervention. Please describe the GCF's financial exit strategy in case of private sector operations (e.g. IPOs, trade sales, etc.).													
The GCF support is required at this stage given the challenges associated to long-term investment in Argentina today (among other, still low – albeit improving – sovereign credit rating), in particular given the financial status of the energy sector. As the government continues to advance on various macroeconomic and sector reforms, these hurdles are expected to be overcome in the mid-term, resulting in a sector that can be financed on a fully commercial basis. Outstanding renewable energy resource conditions in Argentina, adequate availability of other required resources, and the continued decrease in technology cost should ensure financial viability of these investments once the sectorial and general investment challenges are overcome. A more detailed analysis on the pillars for long-term sustainability of GCF target investment and its exit strategy is provided in section D.2.													
E.6.4. Application of best practices													
Please explain how best available technologies and practices are considered and applied. If applicable, specify the innovations/modifications/adjustments that are made based on industry best practices.													
Based on IDBG's vast track-record financing wind and solar PV projects in Latin America, a number of best practices have been identified and consolidated by the IDBG credit team. These practices are considered by project teams and credit officers during due diligence and structuring of these transactions, and implemented as when projects' specific conditions may allow. A number of them are already included in the credit and structuring requirements established for the Program, including those related to financial structuring (ratios, etc.), due diligence review, project contracts, warranties, etc. The other driver and incentive for use of best technologies (beyond any credit requirements IDBG will set as a lender) is the competitive nature of the process, where most efficient, cost effective and reliable technologies (considering the life cycle of plants) will have an advantage over competing options. First during the evaluation of technical proposals (where some may be discarded if not found to be compliant with minimum technical requirements), and secondly, and most importantly, during the evaluation and ranking of economic offers, as price will be the most determinant competitiveness factor of eligible offers and is closely linked with the levelized cost of energy of each project, which captures life cycle costs and performance.													
E.6.5. Key efficiency and effectiveness indicators													
<table border="1"> <tr> <th colspan="2">Estimated cost per t CO₂e, eq. defined as total investment cost / expected lifetime emission reductions (mitigation only)</th> </tr> <tr> <td>(a) Total project financing</td> <td>US\$ 650 Million</td> </tr> <tr> <td>(b) Requested GCF amount</td> <td>US\$ 130 Million (investment component)</td> </tr> <tr> <td>(c) Expected lifetime emission reductions overtime</td> <td>15.3 Million tCO₂e/eq</td> </tr> <tr> <td>(d) Estimated cost per tCO₂e, eq (d = a / c)</td> <td>US\$ 42.4 / tCO₂e/eq</td> </tr> <tr> <td>(e) Estimated GCF cost per tCO₂e, eq (e = b / c)</td> <td>US\$ 8.5 / tCO₂e/eq</td> </tr> </table>		Estimated cost per t CO ₂ e, eq. defined as total investment cost / expected lifetime emission reductions (mitigation only)		(a) Total project financing	US\$ 650 Million	(b) Requested GCF amount	US\$ 130 Million (investment component)	(c) Expected lifetime emission reductions overtime	15.3 Million tCO ₂ e/eq	(d) Estimated cost per tCO ₂ e, eq (d = a / c)	US\$ 42.4 / tCO ₂ e/eq	(e) Estimated GCF cost per tCO ₂ e, eq (e = b / c)	US\$ 8.5 / tCO ₂ e/eq
Estimated cost per t CO ₂ e, eq. defined as total investment cost / expected lifetime emission reductions (mitigation only)													
(a) Total project financing	US\$ 650 Million												
(b) Requested GCF amount	US\$ 130 Million (investment component)												
(c) Expected lifetime emission reductions overtime	15.3 Million tCO ₂ e/eq												
(d) Estimated cost per tCO ₂ e, eq (d = a / c)	US\$ 42.4 / tCO ₂ e/eq												
(e) Estimated GCF cost per tCO ₂ e, eq (e = b / c)	US\$ 8.5 / tCO ₂ e/eq												
Describe the detailed methodology used for calculating the indicators (d) and (e) above.													
GHG emission reductions were calculated based on:													

² While this % is referred to the capacity tendered under the first RenovAr tender, GCF's support will likely meet spread throughout the first few tenders, to optimize its value add.

What's in a Project?

Remember how investment cases we need to create are complex documents with lots of information?

- This means we (may) need to store lots of diverse information in about each project
- Why *may*? Well different funders have different requirements, so not all projects will need all the fields filled

Here are some examples:

01 Project profile

WHAT THE PROJECT IS

Purpose, sector and location
Scope, status and lifecycle
Climate and social baselines
Vulnerabilities and project needs

02 People & governance

WHO IS INVOLVED

Institutions and partner roles
Contributions and responsibilities
Delivery and funding relationships
Governance and oversight

03 Design & delivery

HOW IT WILL BE IMPLEMENTED

Components and activities
Objectives and technical scope
Milestones, dates and completion
Procurement approach and rationale

04 Finance & business model

HOW IT IS FUNDED AND SUSTAINED

Budgets and capital / operating costs
Funding sources, terms and status
Revenue or cost-saving mechanisms
Resources, partners and customers

05 Benefits & co-benefits

WHAT VALUE IT CREATES

Intended benefits and beneficiaries
Environmental and social outcomes
Impact areas and subareas
Connections to measurable indicators

06 Performance & evidence

HOW PROGRESS IS MEASURED

KPI baselines, targets and results
Metrics, units and reporting periods
Source and methodology references
Collection and verification metadata

07 Risk & readiness

WHAT COULD AFFECT DELIVERY

Risk likelihood, impact and mitigation
Monitoring, ownership and status
Permits and certification status
Implementation constraints

08 Policy & alignment

HOW IT FITS THE WIDER CONTEXT

Climate strategies and policy links
Regulatory references and compliance
EU environmental objectives
Do-no-significant-harm assessments

The Tech Stack; Skilling-up

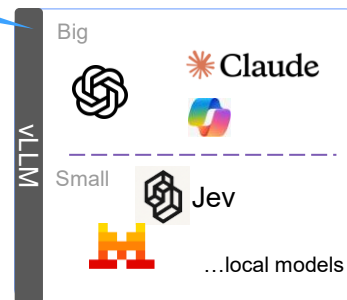
So, we know what data we have, but what processes that data? This is the domain of the second so which contains Tools, Models and the 'Skills' they use

➤ Let's start with AI models

The AI field is moving fast, so there is no good reason to tie yourself to any one model, so put LLM access behind a shared API

Currently, the only distinctions we draw are between big/expensive models (for planning) and cheap/small models (for task completion)

AI Models



DATA



Project Digital Model



Project Search Engine



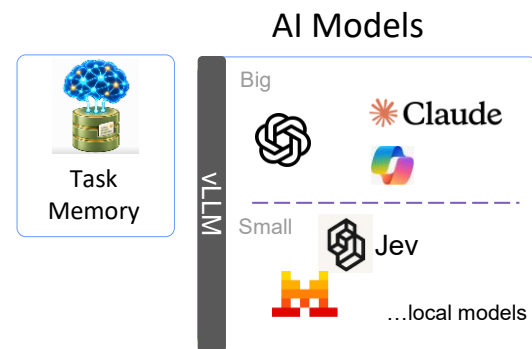
Web Artifact Store



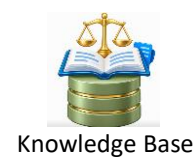
Knowledge Base

The Tech Stack; Skilling-up

Models also don't work without memory, this is done implicitly in a chat environment, but for custom applications we need to handle that ourselves



DATA



The Tech Stack; Skilling-up

As we have an ecosystem of different services that might want to use AI at some point, everyone connecting to the AI models in an ad-hoc manner would be chaotic, so we have a central AI assistant service

Using a common API here allows us to centrally manage resources, and also encode knowledge about how to handle different task types

AI Request
API



AI
Assistant



Task
Memory

AI Models

Big



Claude



VLLM

Small



Jev



...local models

DATA



Project Digital Model



Project Search Engine



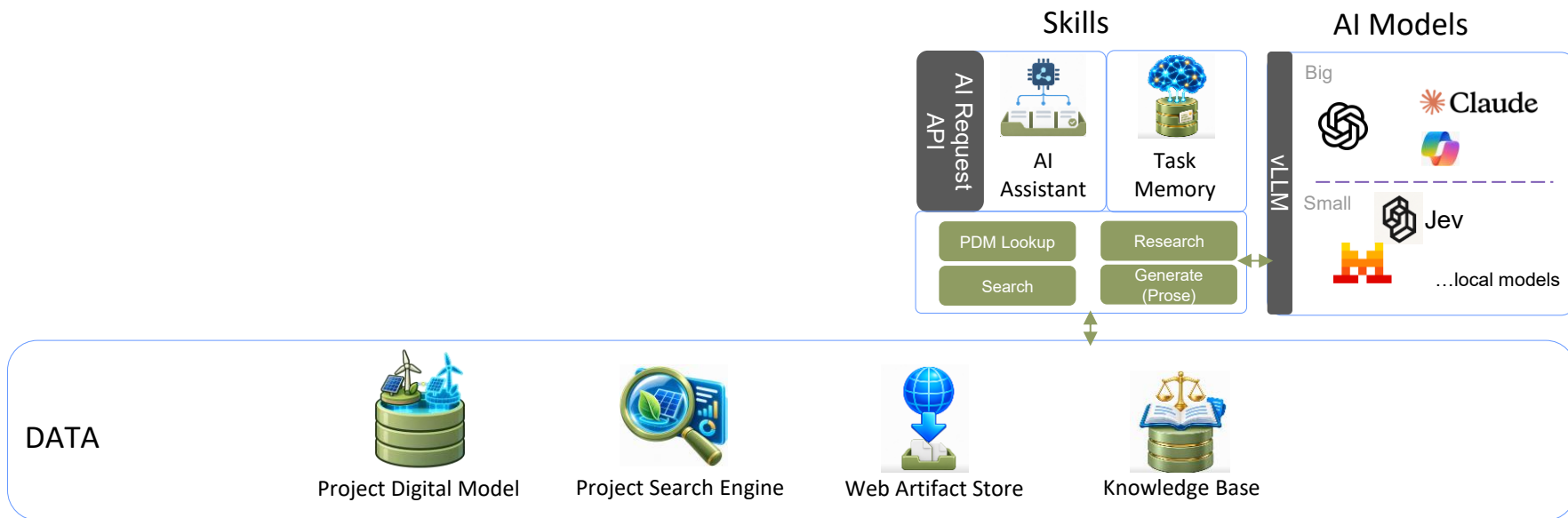
Web Artifact Store



Knowledge Base

The Tech Stack; Skilling-up

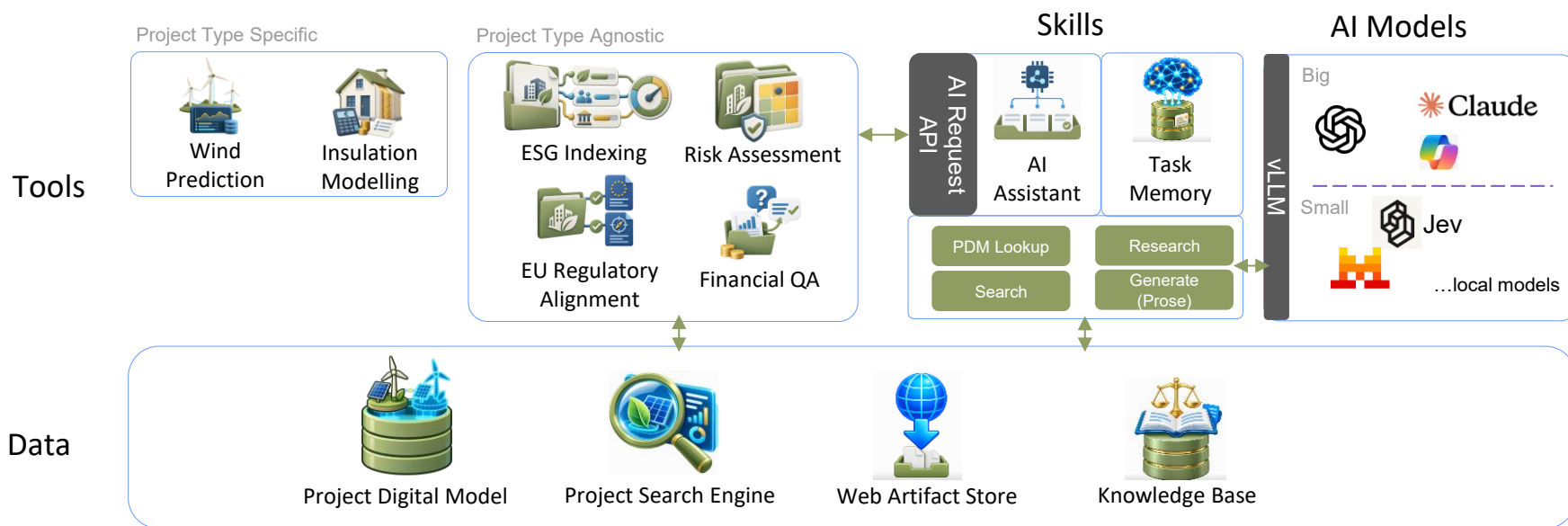
It is also worth including AI skills here, these are pre-registered 'actions' that our assistant can take, practically they are a combination of action definitions and an executor framework (that can run them)



The Tech Stack; Smart Tooling

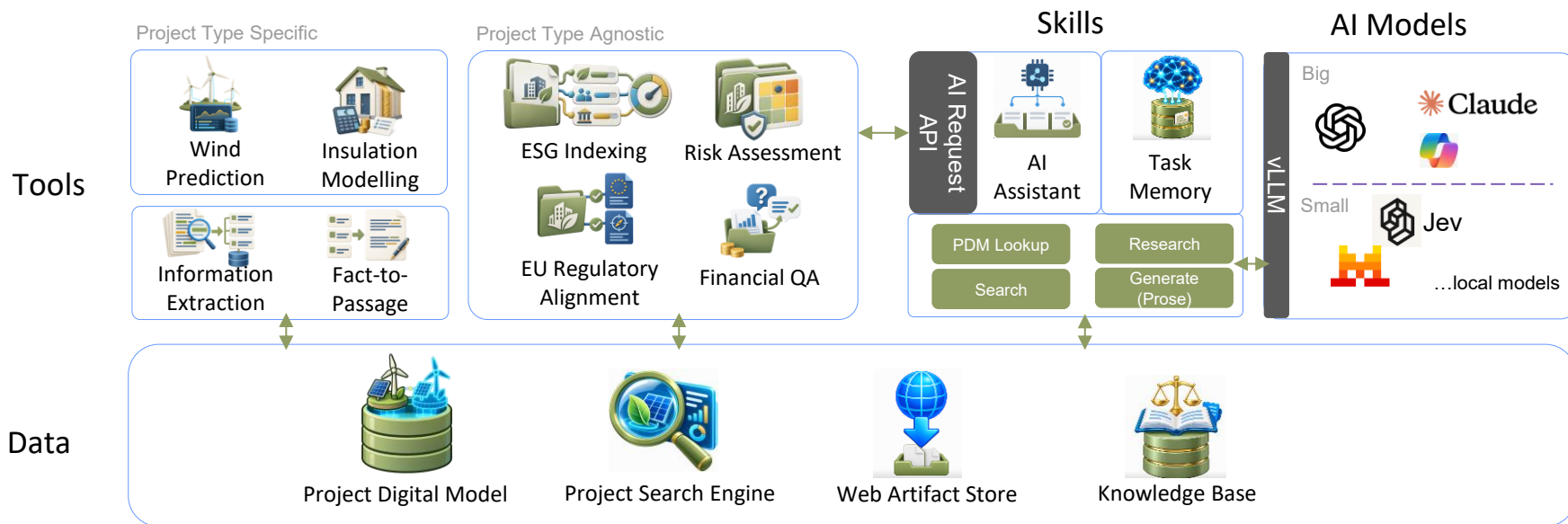
The rest of this layer is comprised of tools that are specific to the Climinvest use-case...

- Tools can be divided based on whether they are relevant to many project types or just one



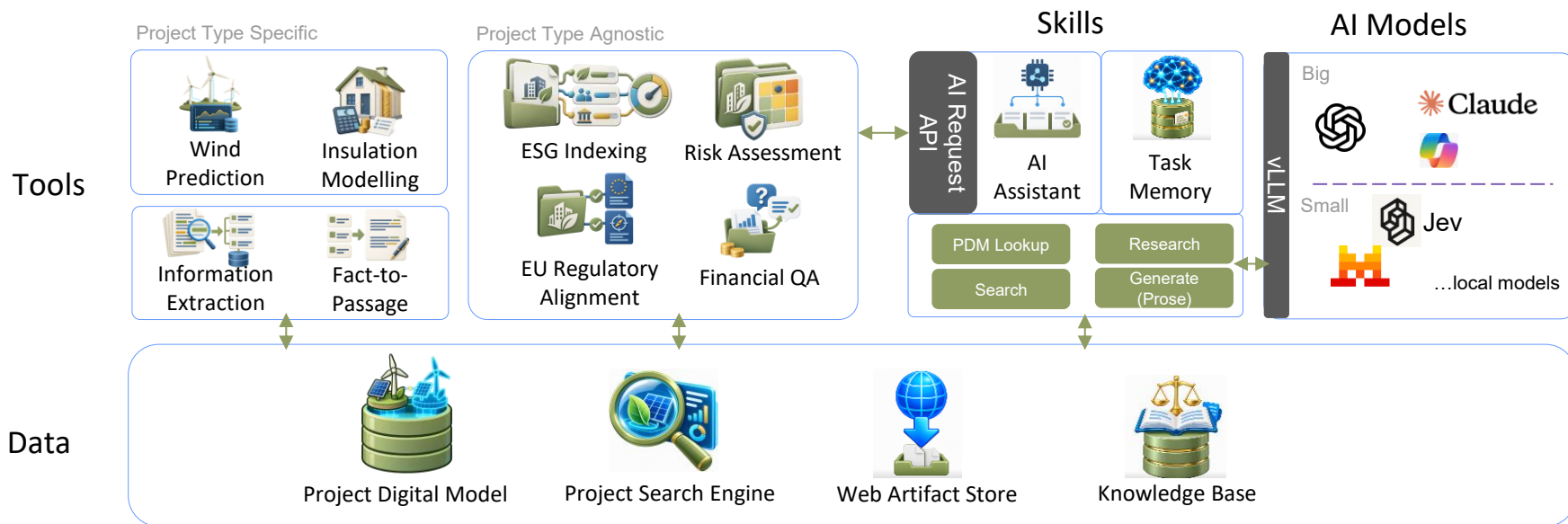
The Tech Stack; Smart Tooling

There are also translation utility tools, which provide some useful automation within the project for converting information from one form to another



The Tech Stack; Web Apps

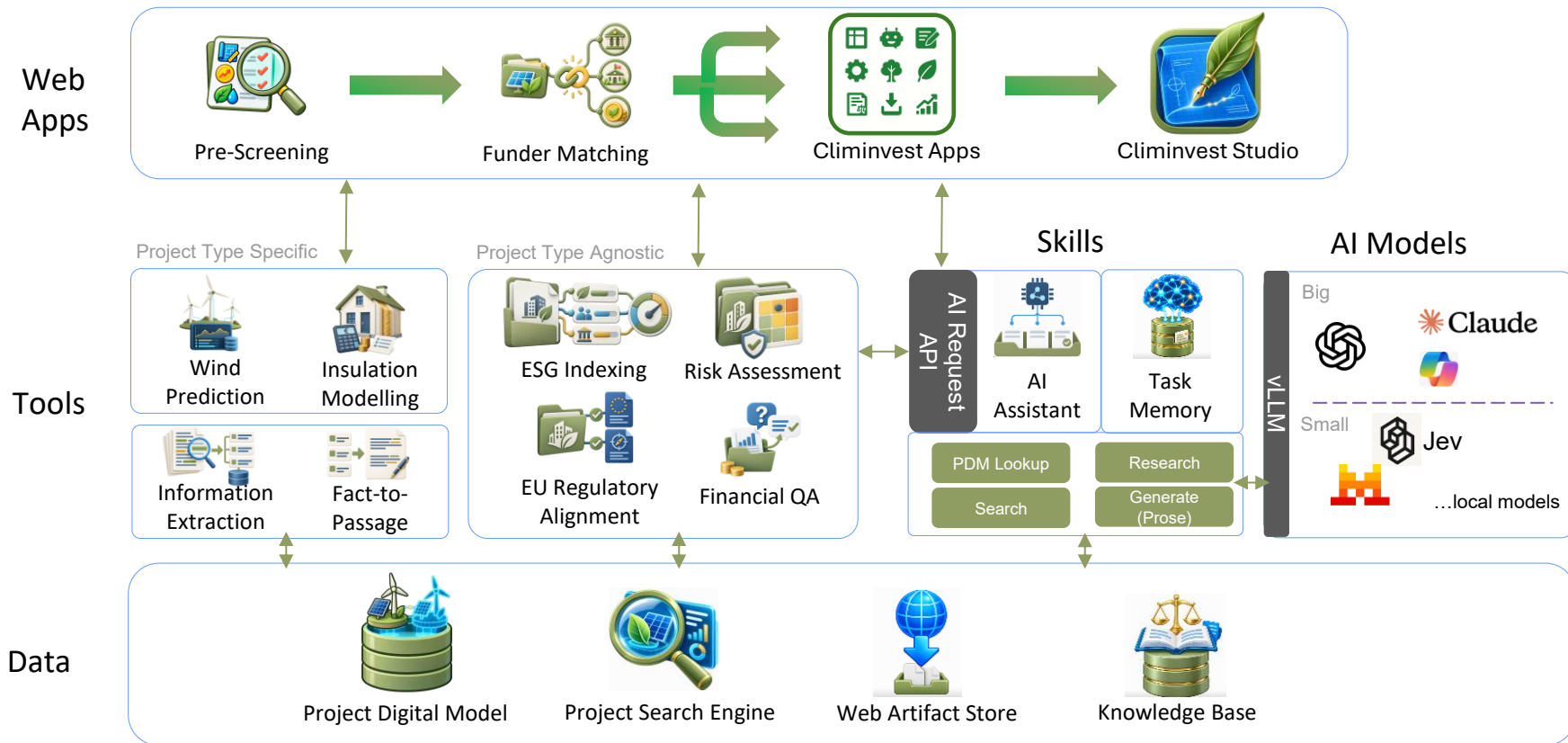
Finally, we get to the bit of the iceberg that the user actually sees!



The Tech Stack; Web Apps



Technology Stack



Agenda

- 09:30 AM - 09:45 AM
 - Welcome & Introduction to Green Investments and the CLIMINVEST Project
- 09:45 AM - 10:15 AM
 - The Bankability-by-Design Framework
- 10:15 AM - 10:35 AM
 - Overview of AI technologies within CLIMINVEST
-  10:35 AM - 10:55 AM
 - Information Extraction from Complex Documents
- 10:55 AM - 11:25 AM
 - Coffee break / Networking



11:25 AM - 11:45 AM

- Financial Question Answering



11:45 AM - 12:05 AM

- Bankability Report Generation

- 12:05 AM - 12:25 PM
 - Demonstration of the CLIMINVEST Studio
- 12:25 PM - 12:45 PM
 - The CLIMINVEST Learning Center & Marketplace
- 12:45 PM - 1:00 PM
 - Future directions and closing